Distribution-Shaped Difficulty Buckets

Chosen geometry, independently checked computational meaning

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The question and the object being represented

Can a chosen distribution place large groups of expensive problems in recognizable regions? We approach this in three steps: learn a score from instance features, reshape its distribution, and independently check the resulting regions. We prove the shape construction, bound the error from a finite reference sample, and certify bucket size and difficulty together. Useful separation requires informative features.

A draw $X \sim D$ is a whole instance sampled from population D. For subset sum, $X = (w, t)$ contains a list of positive integer weights and an integer target. Fix a complete solver A, its implementation, cost unit, and budget $B \geq 0$. Let $C_A(X)$ be its cost and define the budget-exceedance label

$$C_A(X) \geq 0, \qquad H(X) = \mathbf{1}\{C_A(X) > B\}. \tag{1}$$

The indicator $\mathbf{1}\{E\}$ equals one when event E occurs and zero otherwise. Here "hard" means exceeding this solver's budget; it does not imply a NO answer or a lower bound for every algorithm. Running to the budget establishes the label if completion within that budget is observed exactly. A different stopping rule requires its own label.

Extract d inexpensive features and fit a score:

$$\phi(X) \in \mathbb{R}^d, \qquad S = s_\theta(\phi(X)) \in [0, 1].$$

Runtime prediction from instance features is established methodology [1]. Our learner uses logistic regression. For feature j, let $\bar{\phi}_j$ and r_j be its training mean and standard deviation, replacing zero deviations by one. The standardized vector includes an intercept; its coefficient vector is $\theta \in \mathbb{R}^{d+1}$:

$$\widetilde{\phi}(x) = \left(1, \frac{\phi_1(x) - \bar{\phi}_1}{r_1}, \ldots, \frac{\phi_d(x) - \bar{\phi}_d}{r_d}\right),$$

$$s_\theta(\phi(x)) = \operatorname{sigmoid}(\theta^\top \widetilde{\phi}(x)), \qquad \operatorname{sigmoid}(a) = (1 + e^{-a})^{-1}, \tag{2}$$

$$\mathcal{L}(\theta) = -\frac{\sum_{i=1}^{N_{\mathrm{tr}}} v_i [H_i \log S_i + (1 - H_i) \log(1 - S_i)]}{\sum_{i=1}^{N_{\mathrm{tr}}} v_i}. \tag{3}$$

Here N_{tr} is the training size, $H_i = H(X_i)$, and $S_i = s_\theta(\phi(X_i))$. All logarithms are natural. The nonnegative training weights v_i have positive total; they are one by default or clipped observed costs to emphasize expensive cases. The transpose $\theta^\top$ forms the usual dot product. Fixed-step updates use the loss gradient

$$\nabla_\theta \mathcal{L} = \frac{\sum_{i=1}^{N_{\mathrm{tr}}} v_i (S_i - H_i) \widetilde{\phi}(X_i)}{\sum_{i=1}^{N_{\mathrm{tr}}} v_i}.$$

The implementation caps the dot product before evaluating the sigmoid. The guarantees below hold for any fitted score, regardless of its training loss or probability accuracy.

Use three independent samples. *Training* fits the score and standardization; *reference* observations estimate score ranks; *calibration* observations test the frozen regions. For a fixed threshold $\tau \in (0, 1)$, define the predicted group

$$G = \mathbf{1}\{S \geq \tau\}.$$

The group G is distinct from the observed label H. Freeze the predictor, reference transform, buckets, confidence levels, and cost definition before calibration. Repeated targets or scaled copies of one list need a separate independence argument.

Give the representation a chosen distribution

For each group $g \in \{0, 1\}$ of positive probability, let F_g be the conditional cumulative distribution function (CDF):

$$F_g(s) = \mathbb{P}(S \leq s \mid G = g).$$

Choose a continuous target CDF Q_g, strictly increasing on an interval I_g. The two intervals have disjoint, ordered interiors. Draw V uniformly on $[0,1)$, independently of the instance and label, and define

$$U = F_g(S^-) + V\{F_g(S) - F_g(S^-)\}, \qquad Z = Q_g^{-1}(U). \tag{4}$$

Here $F_g(s^-)$ is the left limit at s, and Q_g^{-1} is the target quantile function. In general, the quantile of a CDF F is

$$F^{-1}(u) = \inf\{x : F(x) \geq u\}, \qquad 0 < u < 1.$$

Endpoint values can be assigned by limits. Randomization spreads tied scores across their CDF jump without adding information about difficulty.

A useful choice is a truncated lognormal on $I_g = (\ell_g, u_g)$. Here $0 < \ell_g < u_g$ are the endpoints, and $\mu_g \in \mathbb{R}$ and $\sigma_g > 0$ are location and scale on the logarithmic axis. With Φ the standard normal CDF, put

$$\begin{aligned} a_g &= \Phi((\log \ell_g - \mu_g)/\sigma_g), \quad b_g = \Phi((\log u_g - \mu_g)/\sigma_g), \\ Q_g(z) &= \frac{\Phi((\log z - \mu_g)/\sigma_g) - a_g}{b_g - a_g}, \quad \ell_g < z < u_g, \\ Q_g^{-1}(u) &= \exp\{\mu_g + \sigma_g \Phi^{-1}(a_g + u(b_g - a_g))\}. \end{aligned} \tag{5}$$

Extend the CDF by zero and one outside its interval. A narrow component for the higher-score group creates a compact visible region; calibration will determine whether it actually contains expensive cases.

Theorem 1 (Chosen shape)**.** *Conditional on $G = g$, U is uniform and Z has CDF Q_g. Writing $\pi_g = \mathbb{P}(G = g)$ for the group masses, the full CDF is*

$$\mathbb{P}(Z \leq z) = \pi_0 Q_0(z) + \pi_1 Q_1(z).$$

Proof. Let W be uniform on $[0,1]$. Then $F_g^{-1}(W)$ has CDF F_g. Conditional on this inverse taking a value s of positive probability, W is uniform across the jump from $F_g(s^-)$ to $F_g(s)$. Away from jumps,

$$W = F_g(F_g^{-1}(W)) \quad \text{almost surely.}$$

Equation (4) therefore gives U the same uniform distribution as W. For $z \in I_g$,

$$\mathbb{P}(Z \leq z \mid G = g) = \mathbb{P}(U \leq Q_g(z) \mid G = g) = Q_g(z).$$

Conditioning on the groups gives the mixture. □

The source population fixes the mixture weights. Shaping moves and concentrates each group; changing its population mass requires resampling or reweighting.

In practice, estimate F_g by the empirical CDF $\widehat{F}_g$ of the m_g reference scores in group g. Apply the same jump formula to obtain $\widehat{U}$, then set

$$\widehat{Z} = Q_g^{-1}\big(\mathrm{clip}(\widehat{U}, c, 1-c)\big), \qquad 0 < c < 1/2.$$

The clipping function restricts its first argument to the interval $[c, 1-c]$, avoiding numerical endpoints.

Theorem 2 (Finite-reference shape error)**.** *Let $\eta_g \geq 0$ bound the largest CDF error:*

$$\|\widehat{F}_g - F_g\|_\infty := \sup_s |\widehat{F}_g(s) - F_g(s)| \leq \eta_g.$$

Then

$$\sup_z |\mathbb{P}(\widehat{Z} \leq z \mid G = g, \textit{reference}) - Q_g(z)| \leq \eta_g + c. \tag{6}$$

For a shape failure probability $\delta_s \in (0,1)$ and nonempty reference groups, these bounds hold together with probability at least $1 - \delta_s$ when

$$\eta_g = \min\left\{1, \sqrt{\frac{\log(4/\delta_s)}{2m_g}}\right\}.$$

Proof. The same error bound holds for left limits. Couple true and estimated ranks using the same (S, V). Their difference is a convex combination of the two CDF errors, giving

$$|\widehat{U} - U| \le \eta_g.$$

Clipping adds at most c. A variable within distance e of a uniform variable has CDF within e of the uniform CDF. Use $e = \eta_g + c$, then apply the increasing quantile function.

Conditional on m_g, the group reference scores are independent and identically distributed (i.i.d.) with CDF F_g. For any $\eta > 0$, the DKW–Massart inequality [2] gives

$$\mathbb{P}\big(\|\widehat{F}_g - F_g\|_\infty > \eta \mid m_g\big) \le 2e^{-2m_g\eta^2}.$$

Allocate failure probability $\delta_s/2$ to each group and take a union bound. If the displayed square root exceeds one, use the deterministic error bound one. □

An empty reference group is left unrepresented. The next guarantee applies directly to the implemented frozen buckets, even when the estimated shape is imperfect.

Certify that identifiable buckets are large and hard

Choose a positive integer J, the number of bins per predicted group. Partition the rank range $[0, 1]$ into

$$[j/J, (j+1)/J), \qquad j = 0, \dots, J-1,$$

closing the final endpoint. The group and bin determine a bucket index:

$$b = gJ + j, \qquad b = 0, \dots, M-1, \qquad M = 2J.$$

Write $\mathcal{B}_b$ for the corresponding event in the joint space of (X, V). These buckets are disjoint. Their images under Q_g^{-1} are known coordinate intervals. Choose

$$0 < c < \frac{1}{2J}; \qquad \text{the implementation uses } c = 10^{-9}/J.$$

Clipping then preserves bin membership. Unrepresented groups receive no assignment.

For a fresh draw write

$$\rho_b = \mathbb{P}(\mathcal{B}_b), \qquad p_b = \mathbb{P}(H = 1 \mid \mathcal{B}_b) \quad (\rho_b > 0). \tag{7}$$

Here ρ_b is the bucket's population mass and p_b its hard-case fraction, or purity. Fix permitted impurity $\epsilon \in (0, 1/2)$, minimum mass $\gamma \in (0, 1)$, and calibration failure probability $\delta \in (0, 1)$. Draw N independent calibration observations, each with an independent tie variable V. Let n_b count bucket members and k_b count hard members.

Use the per-test error level α and binomial upper tail T:

$$\begin{aligned} \alpha &= \frac{\delta}{3M}, \\ T(n, k, p) &= \mathbb{P}\{\mathrm{Bin}(n, p) \ge k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}. \end{aligned} \tag{8}$$

Here $\mathrm{Bin}(n, p)$ counts successes in n independent trials with success probability p. Set $T = 1$ for $k \le 0$ and $T = 0$ for $k > n$. Run three tests per bucket:

Claim	Test required to pass
Hard purity $p_b \ge 1 - \epsilon$	$n_b > 0$, $T(n_b, k_b, 1-\epsilon) \le \alpha$
Easy purity $1 - p_b \ge 1 - \epsilon$	$n_b > 0$, $T(n_b, n_b - k_b, 1-\epsilon) \le \alpha$
Mass $\rho_b \ge \gamma$	$T(N, n_b, \gamma) \le \alpha$

Report hard when the hard-purity and mass tests pass, easy when the easy-purity and mass tests pass, and unresolved otherwise. The purity tests cannot both pass. At least one uses a count at most $n_b/2$ with success probability above $1/2$, so its upper tail is at least $1/2 > \alpha$.

Theorem 3 (Simultaneously reliable large buckets)**.** *Fix any training/reference result. With probability at least* $1-\delta$ *over calibration observations and ties, all reported buckets satisfy*

$$\boxed{\begin{array}{ll} \textit{Hard bucket } b: & \rho_b \geq \gamma, \quad p_b \geq 1-\epsilon, \\ \textit{Easy bucket } b: & \rho_b \geq \gamma, \quad p_b \leq \epsilon. \end{array}}$$

Proof. Fix a positive-mass bucket. Conditional on its membership count, the hard count obeys

$$k_b \mid n_b \sim \mathrm{Bin}(n_b, p_b).$$

For a null success threshold $p_0 \in [0,1]$, the counts satisfying $T(n_b, k_b, p_0) \leq \alpha$ form an upper tail of probability at most α under p_0. Coupling trials with common uniforms shows that decreasing the success probability cannot increase rejection. Thus

$$\mathbb{P}\{T(n_b, k_b, p_0) \leq \alpha \mid n_b\} \leq \alpha \quad (p_b \leq p_0).$$

Average over n_b and set $p_0 = 1-\epsilon$ to bound a false hard claim. Apply the same argument to easy counts. For the mass test,

$$n_b \sim \mathrm{Bin}(N, \rho_b),$$

so the tail argument applies at threshold γ. A zero-mass bucket has $n_b = 0$ almost surely and passes no test. A union bound over all $3M$ tests gives

$$\mathbb{P}(\text{any false certification}) \leq 3M\alpha = \delta.$$

The tests need not be independent. This is classical exact-binomial inversion [3] with a shared error budget. □

On this event, let $\mathcal{U}$ be any union of $q \geq 1$ distinct reported hard buckets. Then

$$\mathbb{P}(\mathcal{U}) \geq q\gamma, \qquad \mathbb{P}(H = 1 \mid \mathcal{U}) \geq 1-\epsilon.$$

Disjoint masses add; union purity is their mass-weighted average. The union may be selected after calibration because all candidate buckets were fixed beforehand. A further union bound makes the shape and bucket guarantees jointly valid with probability at least $1 - \delta_s - \delta$.

With defaults $\gamma = 0.05$, $\epsilon = 0.1$, and $\delta = 0.05$, every reported hard bucket contains at least 5% of the population and at least 90% hard cases, at simultaneous confidence 95%. This is a population statement for fresh draws; it is not certainty about an individual case. Calibration is allowed to report no qualifying bucket.

When a hard bucket will be found

Calibration may return no hard bucket. To guarantee finding one, its true mass and purity must exceed the required thresholds. The next result gives a sufficient sample size under those margins.

Theorem 4 (Availability under positive margins)**.** *Suppose a fixed frozen bucket has positive mass and purity margins* $\Delta_\rho, \Delta_p > 0$*:*

$$\rho_b \geq \gamma + \Delta_\rho, \qquad p_b \geq 1 - \epsilon + \Delta_p.$$

Choose an availability failure probability $\beta \in (0,1)$ *and set*

$$L_0 = \max\{\log(1/\alpha), \log(2/\beta)\}.$$

If

$$N \geq 2L_0 \max\left\{\frac{1}{\Delta_\rho^2}, \frac{1}{\gamma\Delta_p^2}\right\}, \tag{9}$$

the calibration algorithm reports that bucket hard with probability at least $1-\beta$*.*

Proof. Hoeffding's bound [4] gives

$$\mathbb{P}\{n_b/N < \gamma + \Delta_\rho/2\} \leq e^{-N\Delta_\rho^2/2}.$$

Outside this event, $n_b \geq N\gamma$ and the mass-test tail satisfies

$$T(N, n_b, \gamma) \leq e^{-N\Delta_\rho^2/2} \leq \alpha$$

by (9). Conditional on any such n_b, Hoeffding also gives

$$\mathbb{P}\{k_b/n_b < 1 - \epsilon + \Delta_p/2 \mid n_b\} \leq e^{-n_b\Delta_p^2/2} \leq e^{-N\gamma\Delta_p^2/2}.$$

Outside the second event, the hard-test tail satisfies

$$T(n_b, k_b, 1-\epsilon) \leq e^{-N\gamma\Delta_p^2/2} \leq \alpha.$$

The sample-size condition bounds each adverse event by $\beta/2$, so both tests pass with probability at least $1 - \beta$. □

Exchange labels for the easy-bucket result. To find all of up to M qualifying buckets with the same confidence, replace $\log(2/\beta)$ by $\log(2M/\beta)$. Exact tests may accept with fewer observations. The margins depend on feature signal in the population; choosing a lognormal shape does not establish them.

Corollary 5 (Aggregate work in a reported region)**.** *Let $\mathbf{1}_{\mathcal{U}}$ indicate membership in a union $\mathcal{U}$ of q reported hard buckets. On the event in Theorem 3, expected cost over a fresh draw satisfies*

$$\mathbb{E}[C_A(X)\mathbf{1}_{\mathcal{U}}] \geq B(1-\epsilon)\mathbb{P}(\mathcal{U}) \geq B(1-\epsilon)q\gamma. \tag{10}$$

Proof. Pointwise,

$$C_A(X)\mathbf{1}_{\mathcal{U}} \geq B\mathbf{1}_{\mathcal{U}\cap\{H=1\}}.$$

Take expectations and use the union's mass and purity bounds. □

This lower-bounds the work assigned to the region. Showing that it contains most total work also requires cost information outside it.

Proposition 6 (What feature information and shaping permit)**.** *Suppose H is independent of $\phi(X)$, and let $p = \mathbb{P}(H = 1)$ be the population hard fraction. Every positive-mass region $\mathcal{R}$ formed from those features and independent randomization satisfies*

$$\mathbb{P}(H = 1 \mid \mathcal{R}) = p.$$

With predictor, reference ranks, and bins fixed, changing the increasing target quantiles preserves bucket membership and purity.

Proof. Condition on the frozen training/reference result. Independence of H from $\phi(X)$ and independent V implies independence from every measurable function of $(\phi(X), V)$. This proves the first statement. For the second, group and rank-bin membership are determined before applying Q_g^{-1}. Replacing that increasing map moves the same members into a different coordinate interval. □

The score supplies cost information, shaping sets the layout, and calibration checks its meaning. Density alone does not certify difficulty.

A complete algorithm and an exact subset-sum application

Offline, fit the score in (2), estimate reference ranks, freeze the M buckets, and run all tests in (8). Online, compute the group, rank, and shaped coordinate, then read the bucket's hard, easy, or unresolved label. Retuning shapes, thresholds, or models on the same calibration labels requires a new error guarantee. Reporting unresolved on unsupported regions follows selective prediction [5].

The implementation uses 12 subset-sum features. Size features measure item count, encoding length, maximum weight bit length, and log normalized total. Target features measure its fraction of the total, distance from a range boundary, and range mismatch. Structural features measure duplicate rate, a greedy superincreasing-core exception rate, log greatest common divisor (gcd), target/gcd mismatch, and normalized maximum weight gap. A superincreasing sequence has each weight larger than the sum of all earlier weights; the core scan keeps weights satisfying that rule and counts rejected items as exceptions. Sorting and linear scans give polynomial bit cost. These features are a starting point for prediction, not a complete description of runtime.

An easy-labeled case proposes exact search through blocks of at most three consecutive sorted weights. Hard-labeled and unresolved cases propose meet in the middle. To make this application standalone, the required block bound is derived next.

Write the sorted weights as $w_1 \le \cdots \le w_n$, with original indices retained, and let L be the binary encoding length of the weights and target. A witness is a set of original indices whose weights sum to the target. For nonempty inputs, choose an integer block limit $1 \le b \le n$ and partition the sorted indices into m nonempty consecutive blocks $B_1, \ldots, B_m$. Define

$$S_j = \left\{ \sum_{i \in W} w_i : W \subseteq B_j \right\}, \qquad P_j = \sum_{h<j} \sum_{i \in B_h} w_i.$$

Thus S_j is the set of distinct local sums, and P_j is the total of all lower blocks. Retain one indexed witness per local sum. The maximum local overlap κ_j and its product K are

$$\kappa_j = \max_{r \in \mathbb{Z}} |S_j \cap [r - P_j, r]|, \qquad K = \prod_{j=1}^{m} \kappa_j. \tag{11}$$

Search blocks from largest to smallest. If higher choices leave residual target r, any feasible local sum s must satisfy

$$0 \le r - s \le P_j \quad \Longleftrightarrow \quad s \in [r - P_j, r].$$

Equal block sums leave the same residual and unprocessed items, so one witness per sum suffices. Recursing on eligible sums is exact by induction on remaining blocks. The level-size bounds are

$$1,\ \kappa_m,\ \kappa_m\kappa_{m-1},\ \ldots,\ K.$$

Each overlap is at least one, so the total is at most $(m+1)K$ nodes.

The partition can also be found efficiently. For each candidate block of indices $[u, v]$ with $1 \le u \le v \le n$ and $v - u + 1 \le b$, construct its distinct-sum set $S_{u,v}$ by sorted-list translation and merging. It contains at most 2^b sums. Its lower total depends only on u, giving the overlap

$$\kappa(u, v) = \max_{r \in \mathbb{Z}} \left| S_{u,v} \cap \left[r - \sum_{i=1}^{u-1} w_i,\ r \right] \right|.$$

A sliding window computes this value from the sorted sums. Let $D(v)$ be the smallest overlap product for a permitted partition of the first v items. Then

$$\begin{aligned} D(0) &= 1, \qquad K_b = D(n), \\ D(v) &= \min_{\max(1, v-b+1) \le u \le v} D(u-1)\kappa(u, v). \end{aligned} \tag{12}$$

Removing the last block and applying induction proves optimality. There are at most nb candidate blocks, each costing $O(2^b)$ arithmetic operations. Sums, witness masks, and products have polynomial bit length. Including discovery, reconstruction, and search gives the bit-time bound

$$\mathrm{poly}(L)(2^b + K_b),$$

where poly denotes a uniform polynomial factor. The empty input is handled directly: only target zero is reachable.

Theorem 7 (Exact router with a general fallback)**.** *On an easy-labeled case, build the representation with blocks of at most three items. Use it only when its optimal overlap product passes the guard*

$$K_3 \le 2^{\lceil n/2 \rceil}.$$

Here K_3 allows blocks of at most three items, including when $n < 3$. Otherwise use meet in the middle; use that fallback directly for hard or unresolved labels. Reject targets satisfying

$$t \notin \left[0, \sum_{i=1}^{n} w_i \right]$$

and verify YES witnesses. Every completed answer is exact, with online bit time at most

$$\mathrm{poly}(L, R)\big(1 + 2^{\lceil n/2 \rceil}\big),$$

where R is the serialized model size in bits.

Proof. Predictions select a complete exact route; they never exclude candidate solutions. The preceding block bound and guard give the claimed exponential scale for that route. In the fallback, build both half-support lists and scan sorted pairs for the target. Each list has at most $2^{\lceil n/2 \rceil}$ entries, and equal sums can retain one witness. This is complete exact matching at the stated bit cost. Feature extraction, inference, and fingerprint checking are polynomial in L, R. □

Let I be the number of gradient iterations and N_{ref} the total reference size. The offline arithmetic-operation bounds are

$$\text{Score fitting:} \quad O(IN_{\mathrm{tr}}d),$$
$$\text{Reference scoring and sorting:} \quad O\big(N_{\mathrm{ref}}d + N_{\mathrm{ref}}\log(N_{\mathrm{ref}}+1)\big).$$

Calibration evaluates rational binomial tails in $O(MN)$ big-integer steps. If the thresholds have encoding length ℓ bits, intermediate integers have $O(N\ell+N)$ bits. Obtaining cost labels is an additional offline expense. Fallback memory is exponential, and bucket purity alone guarantees no speedup.

Verified results and their interpretation

The software uses exact rational confidence tests, fingerprints to detect changes to the frozen model and calibration record, and original-index witness checks. The binomial guarantee applies to the implemented buckets, including floating-point scores and quantiles. The ideal shape results use real arithmetic. Seeded ties are pseudorandom; the probability model assumes independent uniforms.

Verification compares 1020 binomial tails with direct summation, enumerates 240 null rejection bounds, checks 1998 quantile/CDF inversions, and compares 4914 router answers with exhaustive subsets. It also checks ties, empty regions, censored labels, serialization, fingerprints, and all routing labels. These are correctness checks, not performance benchmarks.

A demonstration uses the uniform population $x \in \{0, 0.01, \ldots, 1\}$, feature $\phi(x) = x$, and deliberately constructed costs

$$C(x) = \begin{cases} 1, & x < 0.60, \\ 100, & x \geq 0.60, \end{cases} \qquad B = 10. \tag{13}$$

Training, reference, and calibration independently sample this population with replacement, using 400, 1600, and 2000 observations. The data-draw seed is 20261104; calibration ties use seed 20261105. With two bins per group ($J = 2$), choose

$$(\mu_0, \sigma_0, \ell_0, u_0) = (-1, 0.35, 0.05, 1),$$
$$(\mu_1, \sigma_1, \ell_1, u_1) = (1.6, 0.2, 2, 12).$$

Thus $M = 4$ and the per-test level is

$$\alpha = \frac{0.05}{12} = \frac{1}{240}.$$

Bucket	Label	Calibration count	True mass (%)	True hard fraction (%)
0	Easy	589	28.47	0.00
1	Easy	578	29.95	0.00
2	Hard	399	20.50	95.17
3	Hard	434	21.08	100.00

The true entries use exact integration over randomized ties. For a source score s in group g, write its empirical jump endpoints as

$$a_s = \widehat{F}_g(s^-), \qquad b_s = \widehat{F}_g(s).$$

If $b_s > a_s$, its rank is uniform across this jump. The probability of entering a given bin equals the overlap length divided by $b_s - a_s$. If $b_s = a_s$, the rank is a point. Summing over the 101 equally likely source values gives each bucket's mass and hard fraction.

Both hard buckets exceed 5% mass and 90% purity; their union contains about 41.58% of this synthetic population. Real subset-sum coverage and paid solver improvements require evaluation on a separate untouched test population.

The construction combines classical rank transforms, runtime models, and finite-sample inference. Its novelty remains unestablished; useful coverage requires the feature signal and calibration conditions proved above.

Reproduction. Run `python` on `notes/01-certified-coverage/distribution-shaping/verify.py`. Accompanying `algorithm.py` and `validation.json` supply the method and saved checks. The paper build verifies source hashes and table values without rerunning experiments.

References

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